

Pipeline Provenance & Regeneration Audit — Detailed 4-Phase Output

Date: 2026-03-21

Scope: All figures/tables in `README.md`, `docs/core/*.md`, `docs/analysis/*.md` (excluding `docs/archive/`)

=== 1. PROVENANCE TABLE ===

Technical Report Figures (docs/core/technical_report.md)

Artifact	Location	Generator	Upstream Data	Status
policy_comparison_panel_K20_cap2.png	results/baseline/figures/	scripts/run_production_v2.py	Optimization solver + simulation	✓ Reproducible
re-sponse_time_distribution_by_policy.png	results/baseline/figures/	scripts/run_production_v2.py	all_results_raw.csv	✓ Reproducible
fleet_sensitivity_dual.png	results/archive/figures/	scripts/analysis/regenerate_all_figures.py	all_results_raw.csv	⚠ Archive artifact in active doc
cbd_robustness_enhanced.png	results/analysis/figures/	scripts/analysis/regenerate_all_figures.py	cbd_experiment_results.csv	⚠ Generator has stale read path
queue_comparison_by_policy.png	results/analysis/figures/	scripts/analysis/analyze_queue_metrics.py	Production exp CSVs	⚠ Generator has stale read path
seasonal_patterns.png	results/analysis/figures/	scripts/analysis/analyze_seasonal_patterns.py	crashes_manhattan.csv	⚠ Generator writes to stale output path
distance_comparison_bar.png	results/analysis/distance_comparison/	scripts/analysis/run_distance_comparison_experiment.py	Distance matrices + simulation	✓ Reproducible
cbd_equity_tradeoff_summary.png	results/analysis/figures/	scripts/analysis/regenerate_all_figures.py	cbd_experiment_results.csv	⚠ Generator has stale read path
capacity_sensitivity_heatmap.png	results/archive/figures/	scripts/analysis/generate_capacity_se	simulation_results.csv	● Archive + generator broken (stale read path) +

Artifact	Location	Generator	Upstream Data	Status
		nsitiv- ity_heatmap.py		shows “Data format issue”

Analysis Docs Figures (docs/analysis/*.md)

Artifact	Location	Generator	Upstream Data	Status
performance_vs_capacity_K40.png	results/analysis/capacity_comparison/	scripts/analysis/capacity_sensitivity_analysis.py	Optimization + simulation	✓ Reproducible
tradeoff_dispersions_rt_K40.png	results/analysis/capacity_comparison/	scripts/analysis/capacity_sensitivity_analysis.py	Same	✓ Reproducible
rt_heatmap_K40.png	results/analysis/capacity_comparison/	scripts/analysis/capacity_sensitivity_analysis.py	Same	✓ Reproducible
full_spectrum_summary.png	results/analysis/capacity_comparison/	scripts/analysis/capacity_sensitivity_analysis.py	Same	✓ Reproducible
cbd_response_comparison.png	results/analysis/figures/	notebooks/09_cbd_analysis.ipynb	CBD experiment data	⚠ Notebook-only
cbd_scenario_comparison.png	results/analysis/figures/	notebooks/09_cbd_analysis.ipynb	CBD experiment data	⚠ Notebook-only
cbd_heatmap.png	results/analysis/figures/	notebooks/09_cbd_analysis.ipynb	CBD experiment data	⚠ Notebook-only
queue_comparison_by_policy.png	results/analysis/figures/	scripts/analysis/analyze_queue_metrics.py	Exp CSVs	⚠ Generator has stale read path
queue_vs_fleet_size.png	results/analysis/figures/	scripts/analysis/analyze_queue_metrics.py	Same	⚠ Same
queue_vs_demand.png	results/analysis/figures/	scripts/analysis/analyze_queue_metrics.py	Same	⚠ Same

Artifact	Location	Generator	Upstream Data	Status
queue_heatmap.png	results/analysis/figures/	scripts/analysis/analyze_queue_metrics.py	Same	⚠ Same

Baseline Figures (`results/baseline/figures/`) — via `run_production_v2.py`

Artifact	Location	Generator	Upstream Data	Status
mean_rt_vs_K.png	results/ baseline/figures/	run_production_v2.py	Simulation results	✓ Reproducible
coverage_vs_K.png	results/ baseline/figures/	run_production_v2.py	Simulation results	✓ Reproducible
p95_rt_vs_K.png	results/ baseline/figures/	run_production_v2.py	Simulation results	✓ Reproducible
rt_distribution_K{20,30,40}.png	results/ baseline/figures/	run_production_v2.py	Simulation results	✓ Reproducible
queue_metrics_vs_K.png	results/ baseline/figures/	run_production_v2.py	Simulation results	✓ Reproducible
utilization_vs_K.png	results/ baseline/figures/	run_production_v2.py	Simulation results	✓ Reproducible
effect_sizes.png	results/ baseline/figures/	run_production_v2.py	Simulation results	✓ Reproducible
allocation_map_K{20,30,40}.png	results/ baseline/figures/	run_production_v2.py	Allocation data	✓ Reproducible
exp1_policy_comparison.png	results/ baseline/figures/	run_production_v2.py	Exp 1 results	✓ Reproducible
exp2_fleet_sensitivity.png	results/ baseline/figures/	run_production_v2.py	Exp 2 results	✓ Reproducible
exp3_demand_sensitivity.png	results/ baseline/figures/	run_production_v2.py	Exp 3 results	✓ Reproducible
exp4_service_business.png	results/ baseline/figures/	run_production_v2.py	Exp 4 results	✓ Reproducible

Artifact	Location	Generator	Upstream Data	Status
policy_comparison_panel_K20_cap2.png	results/baseline/figures/	run_production_v2.py	Optimization + simulation	✓ Reproducible
re-sponse_time_distribution_by_policy.png	results/baseline/figures/	run_production_v2.py	all_results_raw.csv	✓ Reproducible
fleet_sensitivity_v2_dual.png	results/baseline/figures/	run_production_v2.py	Simulation results	✓ Reproducible
production_fleet_sensitivity.png	results/baseline/figures/	run_production_v2.py	Simulation results	✓ Reproducible
pub_fig{1..5}_*.png	results/baseline/figures/	run_production_v2.py	Various	✓ Reproducible
project_summary_dashboard.png	results/baseline/figures/	run_production_v2.py	All results	✓ Reproducible
statistical_effect_sizes.png	results/baseline/figures/	run_production_v2.py	Statistical analysis	✓ Reproducible
fig_*.png (EDA figures)	results/baseline/figures/	run_production_v2.py	Crash data	✓ Reproducible
validation_*.png / verification_*.png	results/baseline/figures/	run_production_v2.py	V&V runs	✓ Reproducible
p0_vs_p2_response_time.png	results/baseline/figures/	run_production_v2.py	Policy comparison	✓ Reproducible

Baseline Tables ([results/baseline/tables/](#))

Artifact	Location	Generator	Upstream Data	Status
descriptive_statistics.csv	results/ baseline/ tables/	run_production_v2.py	Simulation results	✓ Reproducible
anova_results.csv	results/ baseline/ tables/	run_production_v2.py	Simulation results	✓ Reproducible
posthoc_comparisons.csv	results/ baseline/ tables/	run_production_v2.py	Simulation results	✓ Reproducible
effect_sizes.csv	results/ baseline/ tables/	run_production_v2.py	Simulation results	✓ Reproducible
confidence_intervals.csv	results/ baseline/ tables/	run_production_v2.py	Simulation results	✓ Reproducible
queue_statistics.csv	results/ baseline/ tables/	run_production_v2.py	Simulation results	✓ Reproducible
table1_baseline_comparison.csv/.tex	results/ baseline/ tables/	run_production_v2.py	Policy results	✓ Reproducible
table2_anova_summary.csv/.tex	results/ baseline/ tables/	run_production_v2.py	ANOVA results	✓ Reproducible
table3_pairwise_comparisons.csv/.tex	results/ baseline/ tables/	run_production_v2.py	Post-hoc tests	✓ Reproducible
table4_sensitivity_summary.csv/.tex	results/ baseline/ tables/	run_production_v2.py	Sensitivity analysis	✓ Reproducible
exp{1..4}_*.csv	results/ baseline/ tables/	run_production_v2.py	Experiment results	✓ Reproducible
production_results.csv	results/ baseline/ tables/	run_production_v2.py	All results	✓ Reproducible

Analysis Artifacts (results/analysis/ subdirs)

Artifact	Location	Generator	Upstream Data	Status
Capacity comparison (21 PNGs + CSVs)	results/analysis/capacity_comparison/	capacity_sensitivity_analysis.py	Optimization + simulation	✅ Reproducible
CBD focused (3 PNGs + CSVs)	results/analysis/cbd_focused_comparison/	run_cbd_focused_optimization.py	CBD experiment data	⚠️ Generator writes to stale path
Distance comparison (4 PNGs + CSVs)	results/analysis/distance_comparison/	run_distance_comparison_experiment.py	Distance matrices + simulation	✅ Reproducible
Allocation maps (3 PNGs)	results/analysis/maps/	run_optimization_comparison.py	Allocation data	⚠️ Generator writes to stale path
Heatmaps (~160 PNGs)	results/analysis/heatmaps/	generate_all_heatmaps.py	Various	✅ Reproducible
CBD analysis figs (6 PNGs)	results/analysis/figures/cbd_*.png	Notebook 09_cbd_analysis.ipynb + regenerate_all_figures.py	CBD experiment data	⚠️ Mixed: notebook-only + stale read paths
Queue figs (5 PNGs)	results/analysis/figures/queue_*.png	analyze_queue_metrics.py	Production CSVs	❌ Broken (stale read path)
Seasonal figs (3 PNGs)	results/analysis/figures/seasonal_*.png	analyze_seasonal_patterns.py	crashes_manhattan.csv	⚠️ Writes to stale output path
P0 spatial figs (3 PNGs)	results/analysis/figures/p0_*.png	p0_spatial_analysis.py	Spatial data	⚠️ Writes to stale output path
Precinct demand figs (3 PNGs)	results/analysis/figures/precinct_*.png	generate_precinct_demand_visualizations.py	demand_lambda_precinct.csv	❌ Broken (wrong column name)

Artifact	Location	Generator	Upstream Data	Status
Tradeoff figs (3 PNGs)	results/analysis/figures/response_time_coverage_*.png	generate_tradeoff_improved.py	Optimization results	⚠ Reads from stale path
Service/travel model figs (5 PNGs)	results/analysis/figures/	Notebook 04_service_travel_proxy.ipynb	Model outputs	⚠ Notebook-only

=== 2. ISSUES FOUND ===

Broken Reproducibility

These scripts **cannot be run at all** — they crash with errors:

1. `generate_capacity_sensitivity_heatmap.py` — `FileNotFoundError`: reads `results/capacity_comparison/`, data is at `results/analysis/capacity_comparison/`
2. `analyze_queue_metrics.py` — No data loaded → crash: reads `results/simulation/production/`, data is at `results/analysis/simulation/production/`
3. `generate_publication_figures.py` — `FileNotFoundError`: reads `results/simulation/production/`, same stale path
4. `generate_summary_dashboard.py` — `FileNotFoundError`: reads `results/tables/` and `results/simulation/production/`, both stale
5. `regenerate_all_figures.py` — Partial (3 of 7 succeed): reads `results/simulation/cbd_experiment/` and `results/capacity_comparison/`, both stale
6. `generate_precinct_demand_visualizations.py` — `KeyError`: `'crash_rate_per_hour'`: column renamed to `lambda_per_hour` in `demand_lambda_precinct.csv`
7. `generate_tradeoff_improved.py` — Would fail: reads `results/optimization/`, data is at `results/archive/optimization/`

Archive Misuse

Archive artifacts referenced in **active** (non-archive) documents:

1. `docs/core/technical_report.md` **line 478** → `results/archive/figures/fleet_sensitivity_dual.png` — Archive figure embedded in active technical report
2. `docs/core/technical_report.md` **line 670** → `results/archive/figures/capacity_sensitivity_heatmap.png` — Archive figure in active report + shows “Data format issue” (blank heatmap)

Stale Scripts

Scripts with stale **INPUT** paths:

Script	Stale Read Path	Correct Read Path
generate_capacity_sensitivity_heatmap.py	results/capacity_comparison/	results/analysis/capacity_comparison/
analyze_queue_metrics.py	results/simulation/production/	results/analysis/simulation/production/
generate_publication_figures.py	results/simulation/production/	results/analysis/simulation/production/
generate_summary_dashboard.py	results/tables/ , results/simulation/production/	results/baseline/tables/ , results/analysis/simulation/production/
regenerate_all_figures.py	results/simulation/cbd_experiment/ , results/capacity_comparison/	results/analysis/simulation/cbd_experiment/ , results/analysis/capacity_comparison/
generate_tradeoff_improved.py	results/optimization/	results/archive/optimization/

Scripts with stale OUTPUT paths:

Script	Stale Output Path	Correct Output Path
<code>analyze_queue_metrics.py</code>	<code>results/figures/</code> , <code>results/tables/</code>	<code>results/analysis/figures/</code> , <code>results/baseline/tables/</code>
<code>ana-lyze_seasonal_patterns.py</code>	<code>results/figures/</code> , <code>results/tables/</code>	<code>results/analysis/figures/</code> , <code>results/baseline/tables/</code>
<code>p0_spatial_analysis.py</code>	<code>results/figures/</code>	<code>results/analysis/figures/</code>
<code>generate_publication_figures.py</code>	<code>results/figures/</code>	<code>results/baseline/figures/</code>
<code>generate_summary_dashboard.py</code>	<code>results/figures/</code>	<code>results/baseline/figures/</code>
<code>regenerate_all_figures.py</code>	<code>results/figures/</code>	<code>results/baseline/figures/</code> or <code>results/analysis/figures/</code>
<code>generate_tradeoff_improved.py</code>	<code>results/figures/</code>	<code>results/analysis/figures/</code>
<code>generate_precinct_demand_visualizations.py</code>	<code>results/figures/</code>	<code>results/analysis/figures/</code>
<code>run_optimization_comparison.py</code>	<code>results/optimization/</code> , <code>results/maps/</code>	<code>results/archive/optimization/</code> , <code>results/analysis/maps/</code>
<code>run_cbd_focused_optimization.py</code>	<code>results/cbd_focused_comparison/</code>	<code>results/analysis/cbd_focused_comparison/</code>

Mismatched Outputs

Script	Column Used	Actual Column in Data	Data File
<code>generate_precinct_demand_visualizations.py</code>	<code>crash_rate_per_hour</code>	<code>lambda_per_hour</code>	<code>demand_lambda_precinct.csv</code>

Additionally, `capacity_sensitivity_heatmap.png` currently shows “Data format issue” (blank panels for K=20 and K=40) — the committed figure does not display valid data.

=== 3. REGENERATION RESULTS ===

Scripts That Run Successfully (paths verified correct)

Artifact / Script	Match Status	Notes
<code>run_production_v2.py</code> → all <code>results/baseline/</code>	Not re-run (810 simulations, long-running)	Paths verified correct; would produce identical structure
<code>capacity_sensitivity_analysis.py</code> → <code>results/analysis/capacity_comparison/</code>	Not re-run (requires simulation)	Paths verified correct
<code>run_distance_comparison_experiment.py</code> → <code>results/analysis/distance_comparison/</code>	Not re-run (requires simulation)	Paths verified correct
<code>run_cbd_experiment.py</code> → <code>results/analysis/simulation/cbd_experiment/</code>	Not re-run (requires simulation)	Paths verified correct
<code>demand_modeling.py</code> → <code>results/baseline/figures/</code>	Not re-run (needs raw crash data)	Paths verified correct
<code>run_verification.py</code> → <code>results/baseline/simulation/verification/</code>	Not re-run	Paths verified correct
<code>run_validation_pilots.py</code> → <code>results/baseline/simulation/validation_pilot/</code>	Not re-run	Paths verified correct
<code>p0_spatial_analysis.py</code>	✅ Ran successfully	Output went to wrong dir (<code>results/figures/</code> instead of <code>results/analysis/figures/</code>)

Scripts That FAIL

Artifact / Script	Match Status	Notes
gener- ate_capacity_sensitivity_he atmap.py	❌ FAIL — FileNotFoundError	Needs path fix: results/capa- city_comparison/ → results/analysis/capa- city_comparison/
analyze_queue_metrics.py	❌ FAIL — No data loaded	Needs path fix: results/simulation/produc- tion/ → results/analysis/ simulation/production/
gener- ate_publication_figures.py	❌ FAIL — FileNotFoundError	Needs same stale simulation path fix
gener- ate_summary_dashboard.py	❌ FAIL — FileNotFoundError	Needs two path fixes (tables + simulation)
regenerate_all_figures.py	⚠️ PARTIAL — 3 of 7 subfunc- tions succeed	CBD + capacity read paths are stale
gener- ate_precinct_demand_visuali- zations.py	❌ FAIL — KeyError	Column rename needed: crash_rate_per_hour → lambda_per_hour
gener- ate_tradeoff_improved.py	❌ Would FAIL	Reads results/optimization/ which is now results/ archive/optimization/

=== 4. RECOMMENDED ACTIONS ===

Keep As-Is

1. All `results/baseline/` **artifacts** — Generated by `run_production_v2.py` with correct paths
2. `results/analysis/capacity_comparison/` **artifacts** — `capacity_sensitivity_analysis.py` paths are correct
3. `results/analysis/distance_comparison/` **artifacts** — `run_distance_comparison_experiment.py` paths correct
4. `results/analysis/simulation/` **artifacts** — Experiment runners have correct paths
5. `demand_modeling.py` — Writes to `results/baseline/figures/` correctly
6. `run_verification.py` and `run_validation_pilots.py` — Both correct
7. `generate_all_heatmaps.py` — Writes to `results/analysis/heatmaps/` correctly

Regenerate and Replace

1. **capacity_sensitivity_heatmap.png** — Currently in archive and shows “Data format issue” (blank panels). After fixing `generate_capacity_sensitivity_heatmap.py`, regenerate to `results/analysis/figures/`
2. **Queue analysis figures (5 PNGs)** — After fixing `analyze_queue_metrics.py` paths, regenerate to `results/analysis/figures/`
3. **Precinct demand figures (3 PNGs)** — After fixing column name in `generate_precinct_demand_visualizations.py`, regenerate to `results/analysis/figures/`
4. **Publication figures** — After fixing `generate_publication_figures.py`, regenerate to confirm match with `results/baseline/figures/` versions

Move to Archive

Artifact	Current Location	Action
No artifacts need moving	—	Current archive contents are already correctly placed

Fix Script (13 scripts need path corrections)

Script	Fix Required
<code>analyze_queue_metrics.py</code>	Read: <code>results/simulation/production/</code> → <code>results/analysis/simulation/production/</code> ; Write figs → <code>results/analysis/figures/</code> ; Write tables → <code>results/baseline/tables/</code>
<code>analyze_seasonal_patterns.py</code>	Write figs → <code>results/analysis/figures/</code> ; Write tables → <code>results/baseline/tables/</code>
<code>generate_capacity_sensitivity_heatmap.py</code>	Read: <code>results/capacity_comparison/</code> → <code>results/analysis/capacity_comparison/</code> ; Write → <code>results/analysis/figures/</code>
<code>generate_publication_figures.py</code>	Read: <code>results/simulation/production/</code> → <code>results/analysis/simulation/production/</code> ; Write → <code>results/baseline/figures/</code> ; Fix CAPACITY=5 → CAPACITY=2
<code>generate_summary_dashboard.py</code>	Read tables: <code>results/tables/</code> → <code>results/baseline/tables/</code> ; Read sim: <code>results/simulation/production/</code> → <code>results/analysis/simulation/production/</code> ; Write → <code>results/baseline/figures/</code>
<code>regenerate_all_figures.py</code>	Read CBD: → <code>results/analysis/simulation/cbd_experiment/</code> ; Read capacity: → <code>results/analysis/capacity_comparison/</code> ; Write: split baseline/analysis
<code>generate_tradeoff_improved.py</code>	Read: <code>results/optimization/</code> → <code>results/archive/optimization/</code> ; Write → <code>results/analysis/figures/</code>
<code>generate_precinct_demand_visualizations.py</code>	Fix column: <code>crash_rate_per_hour</code> → <code>lambda_per_hour</code> ; Write → <code>results/analysis/figures/</code>
<code>p0_spatial_analysis.py</code>	Write: <code>results/figures/</code> → <code>results/analysis/figures/</code>
<code>run_optimization_comparison.py</code>	Write results: → <code>results/archive/optimization/</code> ; Write maps: → <code>results/analysis/maps/</code>
<code>run_cbd_focused_optimization.py</code>	Write: → <code>results/analysis/cbd_focused_comparison/</code>

Fix Technical Report References

Location	Current Reference	Recommended Fix
docs/core/technical_report.md line 478	results/archive/figures/fleet_sensitivity_dual.png	Replace with results/baseline/figures/fleet_sensitivity_v2_dual.png
docs/core/technical_report.md line 670	results/archive/figures/capacity_sensitivity_heatmap.png	Regenerate to results/analysis/figures/capacity_sensitivity_heatmap.png and update reference

Update Documentation Indexes

Document	Issue
docs/core/visualization_index.md	~100+ references to results/figures/ → update to results/baseline/figures/ or results/analysis/figures/
docs/core/figure_trace_guide.md	Same — all locations reference stale results/figures/
docs/core/data_usage_guide.md	References results/figures/*.png generically; update to reorganized paths

Summary Statistics

Category	Count
Total artifacts audited	~120 figures + ~40 tables
Fully reproducible (script + correct paths)	~75 figures, ~30 tables
Broken scripts (won't run)	7 scripts
Scripts with stale output paths	10 scripts
Scripts with stale input paths	6 scripts
Archive artifacts in active docs	2 (technical report)
Notebook-only figures (no script)	15 figures
Column name mismatches	1 (<code>crash_rate_per_hour</code> vs <code>lambda_per_hour</code>)

Root Cause

All issues stem from **a single organizational change**: the directory reorganization from flat `results/figures/`, `results/tables/`, `results/simulation/`, `results/optimization/`, `results/capacity_comparison/` into the tiered `results/baseline/`, `results/analysis/`, `results/archive/` structure. **The scripts were not updated to match the new directory structure.** The data and figures themselves are valid; only the paths in the generating scripts are stale.

Priority Ranking

1. ● **P0 — Fix the 7 broken scripts** (can't regenerate at all)
2. ● **P1 — Fix technical report archive refs** (TR lines 478, 670)
3. ● **P2 — Fix 10 scripts with stale output paths** (would write to wrong dirs)
4. ● **P3 — Update `visualization_index.md` and `figure_trace_guide.md`** (informational)
5. ● **P4 — Create script equivalents for notebook-only figures** (nice-to-have)